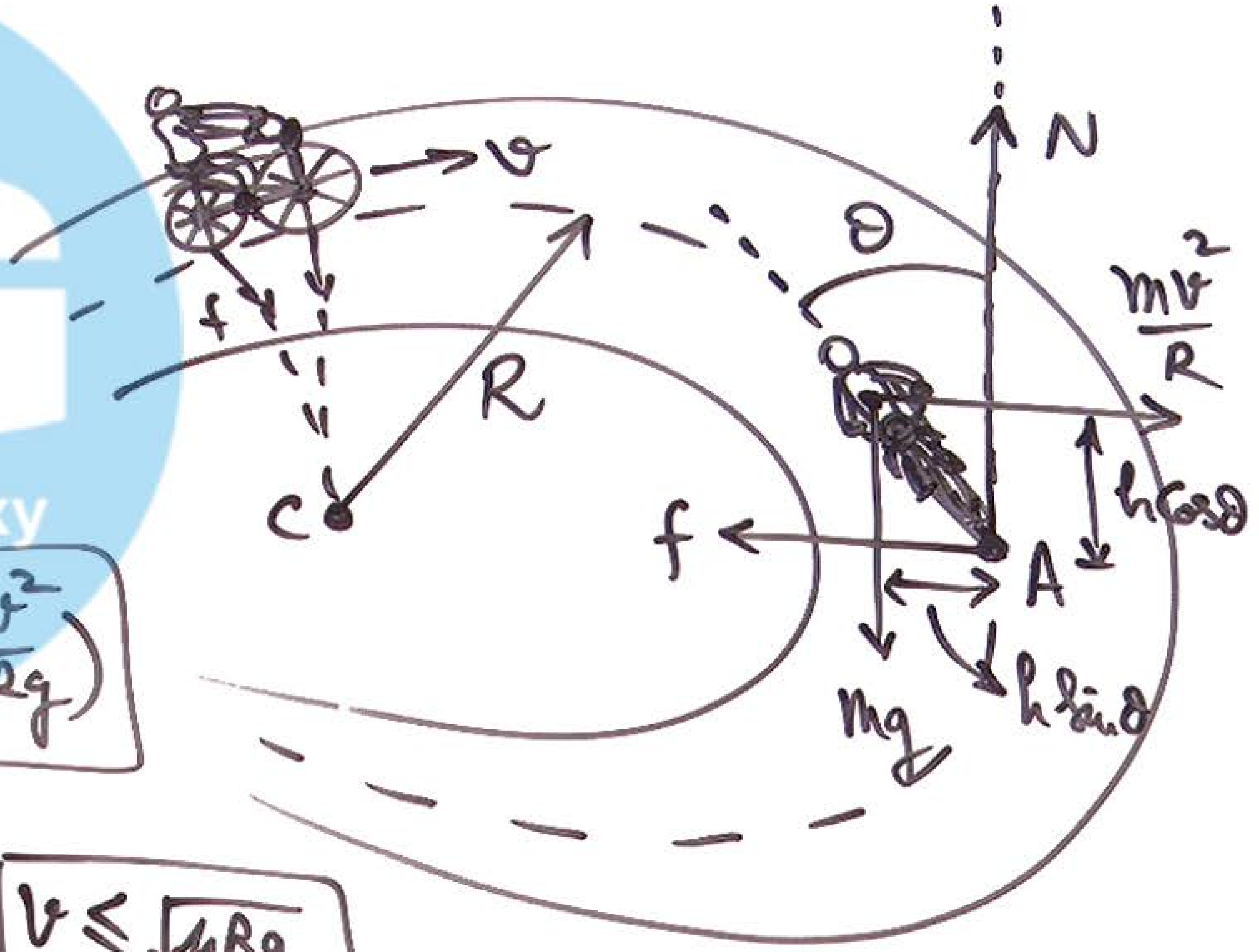
(1) Turning of a 2-wheeler on a flat Road :

for safe turning of bike without toppling

$$\frac{mv^2}{R}\cdot h\cos\theta = mg\cdot h\sin\theta$$

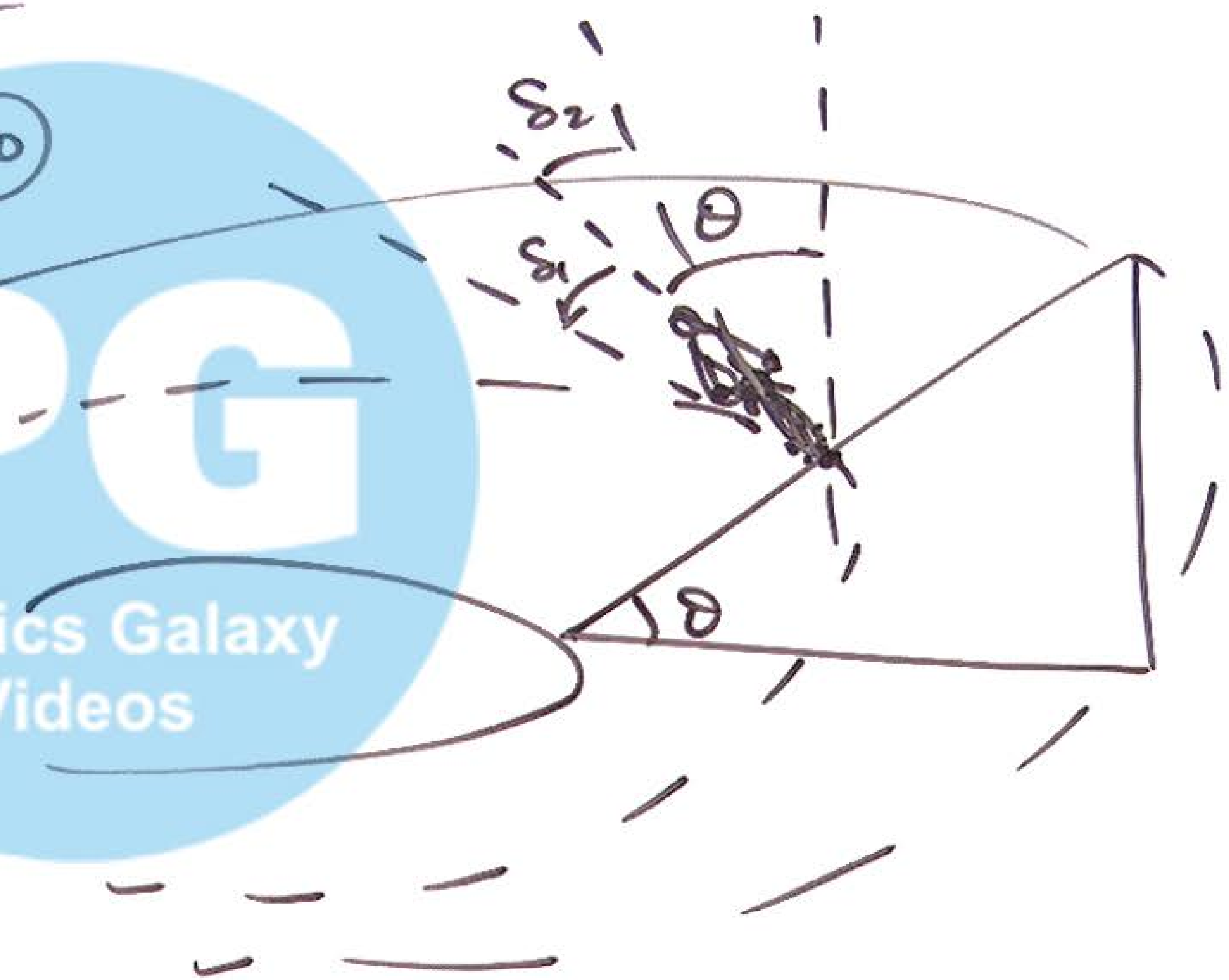
$$\Rightarrow \quad \tan\theta = \frac{v^2}{Rg} \Rightarrow \boxed{\theta = \tan^{-1}\left(\frac{v^2}{Rg}\right)}$$



for safe turning without skidding

$$\frac{mv^2}{R} \le f_L = \mu(mg) \Rightarrow \boxed{v \le \sqrt{\mu Rg}}$$

② Turning of a 2-wheeler on Banked Road:



$$\theta = \tan^{-1}\left(\frac{v^2}{Rg}\right)$$

If $v_1 > v_{sl}$ then angle of tilt

$$\theta_1 = \tan^{-1}\left(\frac{v_1^2}{Rg}\right) > \theta$$

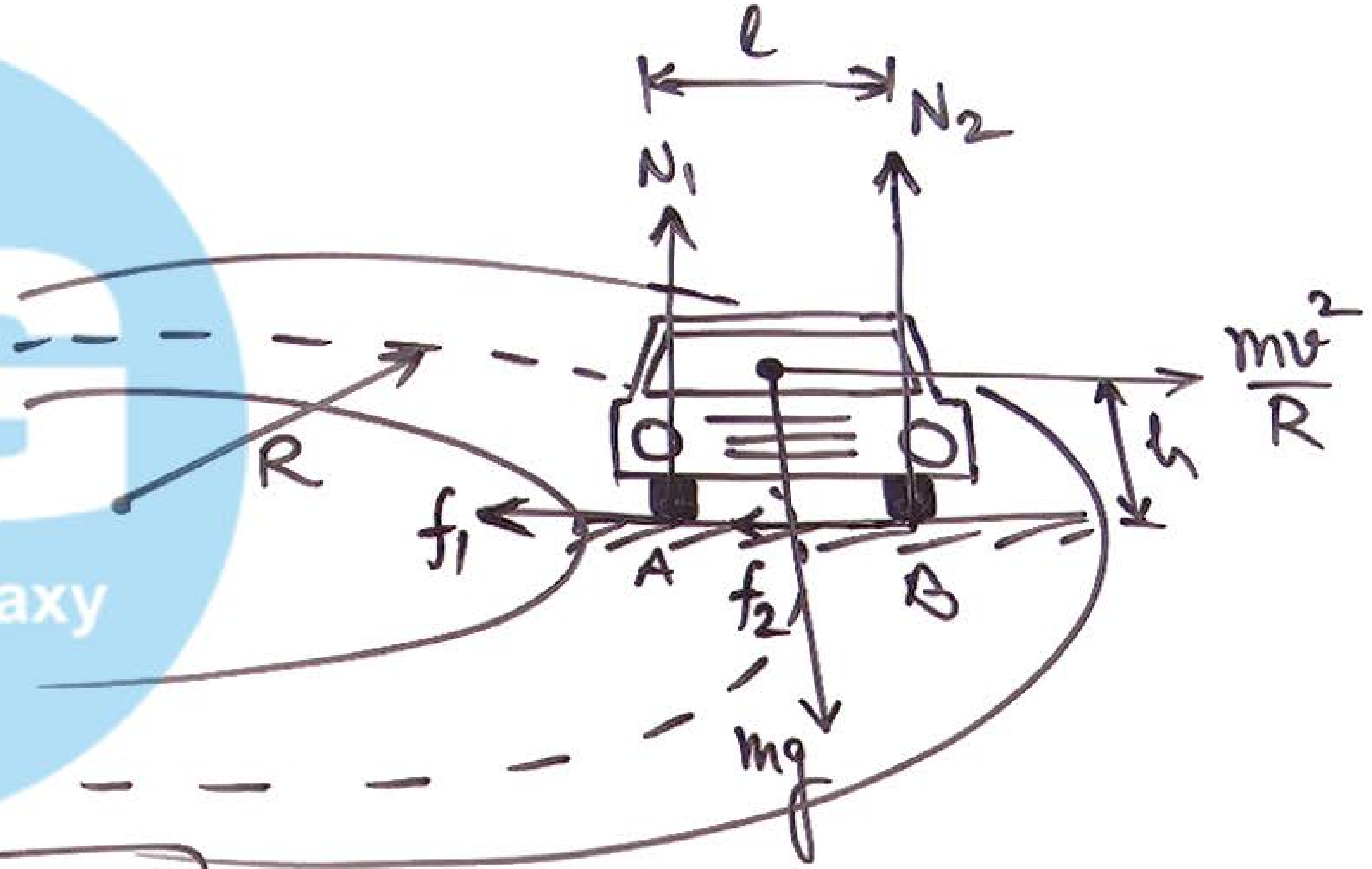
$$\delta_1 = \theta_1 - \theta$$

$$\delta_2 = \theta - \theta_2 \text{ where } \theta_2 = \tan^{-1}\left(\frac{v_2^2}{Rg}\right) < \theta$$

③ Turning of 4-Wheelers on a flat Road:

for safe turning without skidding

$$\boxed{v \le \sqrt{\mu R g}}$$



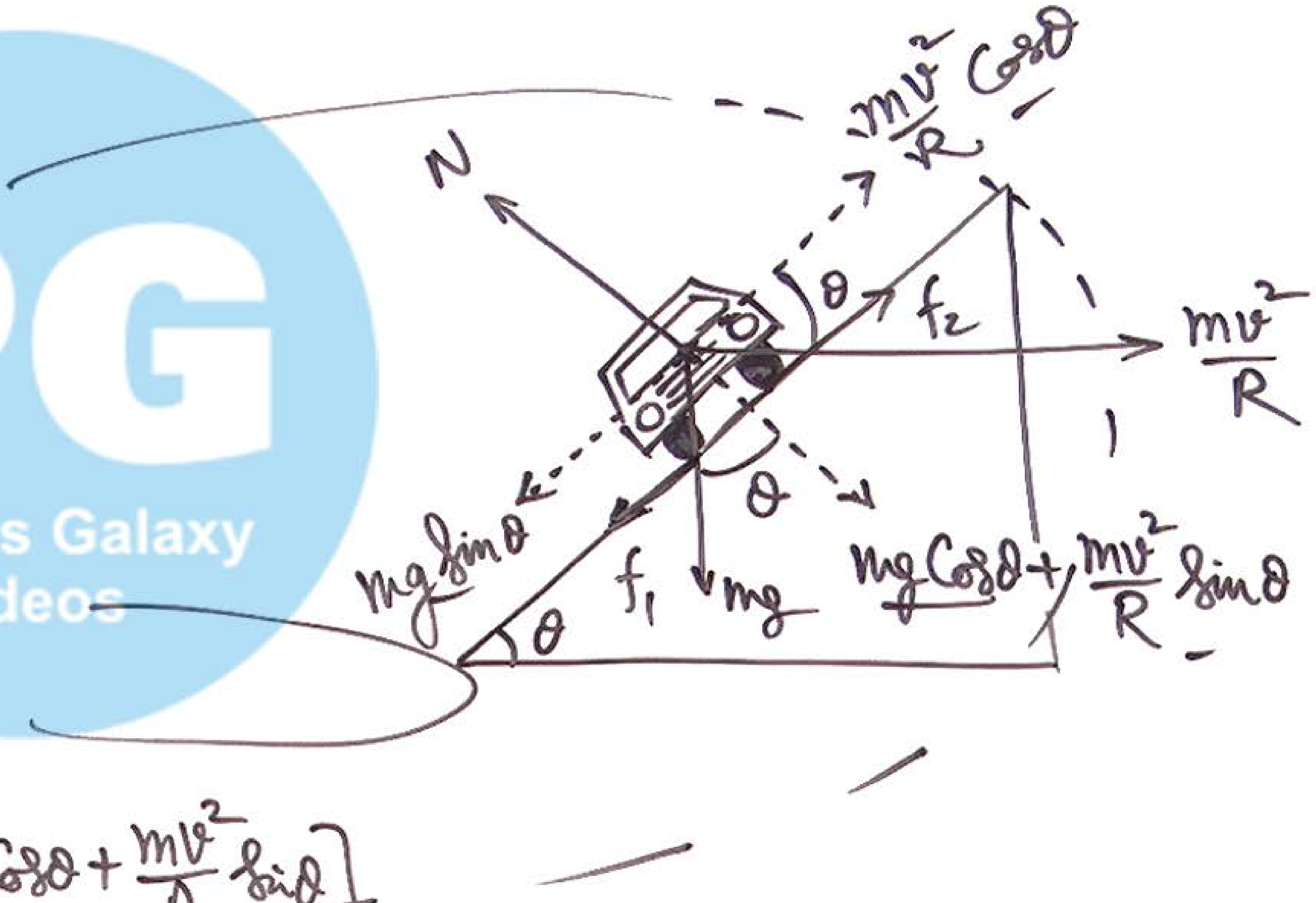
for safe turning without overturning

$$\cancel{m}g \times \frac{l}{2} \ge \frac{\cancel{m}v^2}{R} \times h \Rightarrow \boxed{v \le \sqrt{\frac{Rgl}{2h}}} \Rightarrow \text{for } N_1 = 0.$$

(4) Turning of 4-wheeler on a Banked Road:

for a speed limit v, turning track should be banked at

$$\theta = \tan^{-1}\left(\frac{v^2}{Rg}\right)$$



Condition of outward skidding

$$\frac{mv^2}{R}\cos\theta - mg\sin\theta \geq \mu\left[mg\cos\theta + \frac{mv^2}{R}\sin\theta\right]$$

(5) Particle inside a Rotating Hemisphere :

for no sliding tendency of particle,

we use

$$mg\sin\theta = m\omega^2 r\cos\theta$$

$$\boxed{\tan\theta = \frac{\omega^2 r}{g}} \Rightarrow \boxed{\cos\theta = \frac{g}{\omega^2 R}}$$